Same Planet, Three Codes: Inter-Code Systematics in Exoplanet Transmission Spectra and Their Effect on Machine-Learning Retrievals

Draft — working title (option 1 of 5)

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Abstract

Interpreting exoplanet transmission spectra requires a radiative-transfer forward model, and machine-learning retrievals inherit the one used to build their training set. We quantify how much this choice matters with a paired experiment. We compute 98,725 isothermal, cloud-free spectra with each of POSEIDON, petitRADTRANS, and TauREx from identical parameters, binned onto a common 52-bin Ariel grid (0.55–7.28 μm). Among the 43,885 planets with features of at least 210 ppm, POSEIDON spectra are 21–24 ppm shallower in the median than those of the other two codes. After removing this offset, the codes differ by a median of 8–10 ppm (1.0–1.3% of the feature height). The four leading principal components of these differences are predicted from the planet parameters with out-of-sample $R^2 \geq 0.94$. One-class autoencoders trained on a single code identify spectra from another code with AUC = 0.962–0.992 without noise. At 10 ppm of white noise per bin, the AUC falls to 0.628, or 0.735 with an error-weighted χ^2 score. At a calibrated 1% false-positive rate the autoencoders detect 4.1% of other-code spectra between 10 and 50 ppm, against 36–81% for an oracle that knows the true parameters. A supervised random forest reaches a pairwise AUC of 0.945 at 10 ppm (+0.210 over the best autoencoder score) and 0.775 at 50 ppm (+0.179). Applying gradient-boosted retrievals trained on one code to spectra from another lowers R^2 by 0.224 for the CO abundance and 0.172 for the planet radius. It also biases the temperature by up to 45 K in the median and by -330 K for the hottest 5% of planets. Increasing the training set twenty-fold doubles the self/cross gap for CO, from 0.106 to 0.224. Under noise these biases persist while the scatter grows, so they matter little for individual planets observed at $\gtrsim 30$ ppm precision but remain a coherent shift for population studies.

1 Introduction

Transmission spectroscopy constrains the composition and thermal structure of exoplanet atmospheres by measuring the wavelength dependence of the transit depth [TODO_seager_sasselov_2000_transmission]. The James Webb Space Telescope now delivers transmission spectra with per-bin precision of tens of parts per million [TODO_jwst_ers_transiting_2023], and the Ariel mission will observe the atmospheres of about a thousand planets with a common instrument and a common data-reduction pipeline [TODO_tinetti_ariel_2018]. Every physical interpretation of these data passes through a forward model, a radiative-transfer code that maps atmospheric parameters to a spectrum. Several independent codes are in common use, including POSEIDON [TODO_macdonald_poseidon], petitRADTRANS [TODO_molliere_petitradtrans], and

TauREx [TODO_alrefaie_taurex3]. These codes differ in opacity data, opacity treatment (line-by-line cross-sections or correlated-k tables), reference-pressure conventions, and numerical integration schemes. Machine-learning retrievals, trained on large grids of simulated spectra, are increasingly used to amortise the cost of Bayesian inference [TODO_marquez_neila_2018_hela,TODO_ml_retrieval_review]. Each such retrieval inherits the forward model used to generate its training set.

How much the choice of forward model matters remains poorly quantified at the population level. Published inter-code comparisons have mostly examined retrieval outcomes on a small number of benchmark spectra, or forward-model agreement for a few reference atmospheres [TODO_barstow_2020_retrieval_comparison,TODO_baudino_2017_model_comparison][TODO: VERIFY this characterisation of the literature]. Three questions remain open. First, what are the amplitude, spectral shape, and parameter dependence of the differences between codes across a broad prior? Second, can a single observed spectrum reveal which code best describes it, given realistic noise? Third, how large are the biases incurred when a data-driven retrieval trained on one code is applied to spectra from another? The third question bears directly on ML retrievals, which are usually trained on one code and validated on spectra from that same code [TODO_ml_retrieval_single_code_validation], so their reported accuracy contains no forward-model uncertainty.

We answer these questions with a paired experiment. We compute 98,725 cloud-free, isothermal transmission spectra with each of POSEIDON, petitRADTRANS, and TauREx, using identical parameters for every planet, and bin them onto a common 52-bin Ariel grid from 0.55 to 7.28 μm. Because every planet exists in all three codes, any difference between the codes' spectra or retrievals can be attributed to the code alone. We decompose the paired differences into an offset and a shape residual and identify their spectral structure with principal component analysis. We measure detectability with one-class autoencoders trained on a single code and compare them with a supervised three-class classifier and an oracle that knows the true parameters. Finally, we quantify the retrieval error caused by code mismatch with gradient-boosted regressors, as a function of training-set size and observational noise.

Our contributions are as follows.

- **A population-scale map of inter-code systematics.** Over 43,885 planets with features of at least 210 ppm, POSEIDON spectra are shallower than those of petitRADTRANS and TauREx by a median of 21–24 ppm, and this offset scales with feature height (Spearman $\rho_S = -0.89$ to -0.96). After the offset is removed, the codes differ by a median of 8–10 ppm (1.0–1.3% of the feature height), and by 75–91 ppm at the 90th percentile. The leading four principal components of the relative residuals are predicted from the nine planet parameters with out-of-sample $R^2 \geq 0.94$, so the systematics are low-dimensional and nearly deterministic.

- **Detectability limits from a single spectrum.** Noise-free one-class autoencoders identify the generating code with AUC = 0.962–0.992. Under white noise of 10–50 ppm per bin, at a calibrated 1% false-positive rate, they flag only 4.1–4.2% of other-code spectra, while an oracle that knows the true parameters flags 36–81%. A supervised random forest reaches a pairwise AUC of 0.945, 0.886, and 0.775 at 10, 20, and 50 ppm, against 0.735, 0.669, and 0.596 for the best autoencoder score. The limiting factor is the autoencoders' rare large reconstruction errors (99th percentile 85–135 ppm), not the information content of the data.

- **Retrieval bias from code mismatch, which more training data does not remove.** Applying a retrieval trained on one code to spectra from another lowers R^2 by up to 0.224

(CO abundance) and 0.172 (planet radius). It also introduces median temperature biases of up to 45 K, which reach −330 K for the hottest 5% of planets. Increasing the training set from 5% to 100% doubles the self/cross gap for CO, from 0.106 to 0.224, because a more accurate emulator of one code is a more specialised one.

- **The interplay of noise, bias, and planet type.** The code-induced biases stay nearly constant under noise while the scatter grows: the median bias-to-scatter ratio falls from 0.10 without noise to 0.08 at 50 ppm, and the ratio for temperature reaches 0.72 without noise and 0.39 at 50 ppm. For individual planets the systematics are therefore secondary beyond about 30 ppm, but for population studies they remain a coherent shift. The same cool and low-gravity planets are the hardest to reconstruct in all three codes (median T of 770 K, against 1940 K for the remaining planets): POSEIDON and petitRADTRANS share 479 of their 658 worst-reconstructed planets, against 33 expected by chance. Under noise, the low-gravity planets are also where the code differences are most detectable.

We release the paired spectra, the trained models, and the analysis code [TODO: VERIFY repository URL and archive DOI].

Section 2 reviews related work. Section 3 formalises the problem and the methods, and Section 4 describes the experimental setup. Section 5 presents the results, and Section 7 discusses limitations and concludes.

2 Related Work

Forward models for transmission spectra. Transmission spectra are computed by integrating the slant optical depth through a model atmosphere, given a pressure–temperature profile, a chemical composition, and opacity data [TODO_brown_2001_transmission_model, TODO_benneke_seager_transmission_scaleheight]. Open-source codes implement this calculation with different design choices. POSEIDON [TODO_macdonald_poseidon] and TauREx [TODO_alrefaie_taurex3] compute spectra from cross-sections tabulated at high resolution. petitRADTRANS [TODO_molliere_petitradtrans] offers both line-by-line and correlated-k modes. Other widely used frameworks include NEMESIS, CHIMERA, PLATON, and ATMO [TODO_other_forward_model_codes]. The analytic dependence of the transit depth on the reference pressure and the scale height [TODO_heng_kitzmann_2017_reference_pressure]is directly relevant to the depth offset we find between POSEIDON and the other two codes.

Inter-code comparisons and opacity uncertainty. Disagreements between codes are recognised as a source of systematic error in atmospheric characterisation. Comparisons of retrieval frameworks on shared synthetic spectra [TODO_barstow_2020_retrieval_comparison] and of forward models for JWST-quality data [TODO_baudino_2017_model_comparison] have shown that codes can disagree beyond the expected observational uncertainties. Uncertainties in line lists, broadening parameters, and continuum opacities propagate into retrieved abundances [TODO_niraula_2022_opacity_challenge]. The correlated-k approximation introduces its own band-dependent errors relative to line-by-line calculations [TODO_lacis_oinas_1991_correlated_k,TODO_correlated_k_vs_line_by_line_accuracy]. These studies have typically examined a small number of reference atmospheres [TODO: VERIFY]. Our paired design measures the full distribution of inter-code differences over a nine-dimensional prior, relates each component of that distribution to the planet parameters, and connects it quantitatively to detectability and retrieval bias.

Machine-learning retrievals. Supervised regression from spectra to parameters was introduced with random forests [TODO_marquez_neila_2018_hela] and extended to deep networks [TODO_cobb_2019_plannet,TODO_yip_2021_interpretable_dl_retrieval], generative models [TODO_zingales_waldmann_2018_exogan], normalising flows [TODO_ardevol_martinez_2022_flows,TODO_vasist_2023_npe], and community benchmarks for Ariel [TODO_ariel_data_challenge]. These works train and test on spectra from the same forward model, and therefore measure the emulation error of a retrieval but not its sensitivity to the forward model. We measure that sensitivity directly, and show that the self/cross gap grows with training-set size.

Model misspecification in simulation-based inference. Amortised and simulation-based inference relies on the simulator being a faithful description of the data [TODO_cranmer_2020_sbi_frontier]. When it is not, posterior estimators can be confidently wrong [TODO_hermans_2022_sbi_trust_crisis], which has motivated methods that detect misspecification from summary statistics or learned embeddings [TODO_schmitt_2023_misspecification_detection]. The machine-learning literature treats the same problem as dataset shift [TODO_quinonero_candela_2009_dataset_shift,TODO_ben_david_2010_domain_adaptation]. Code mismatch is a concrete, physically motivated instance of simulator misspecification. Here the misspecification is fully known, because the alternative simulators are available, which allows the misspecification bias to be measured exactly.

Autoencoders and one-class anomaly detection. Reconstruction-based anomaly detection with autoencoders assigns high scores to inputs that lie off the learned data manifold [TODO_sakurada_yairi_2014_ae_anomaly,TODO_ruff_2021_unifying_anomaly_review]. Denoising autoencoders learn representations that are robust to input corruption [TODO_vincent_denoising_ae]. In astronomy, autoencoders have been used to compress and model galaxy and stellar spectra and to flag unusual objects [TODO_portillo_2020_sdss_vae, TODO_astro_spectra_anomaly_detection]. Reconstruction-based detectors are sensitive to the tails of the nominal error distribution, and are generally preferred when the anomalous class is unknown, whereas supervised detectors are preferable when examples of it are available [TODO_one_class_vs_supervised_tradeoff]. Our results confirm both effects in a physical setting: the nominal error tail sets the detection limit under noise, and a classifier given examples of the alternative codes performs far better.

3 Methodology

3.1 Problem definition

Let $\boldsymbol{\theta} \in \Theta \subset \mathbb{R}^9$ denote the parameters of a transiting planet,

$$\boldsymbol{\theta} = (R_\star,\ R_p,\ M_p,\ T,\ \log_{10} X_{\mathrm{H_2O}},\ \log_{10} X_{\mathrm{CH_4}},\ \log_{10} X_{\mathrm{CO}},\ \log_{10} X_{\mathrm{CO_2}},\ \log_{10} X_{\mathrm{NH_3}}), \tag{1}$$

where $R_\star$ is the stellar radius, R_p and M_p are the planetary radius and mass, T is the temperature of an isothermal atmosphere, and X_s is the volume mixing ratio of species s. A radiative-transfer code c defines a forward model

$$\mathcal{F}^{(c)} : \Theta \to \mathbb{R}^D, \qquad \mathbf{y}^{(c)} = \mathcal{F}^{(c)}(\boldsymbol{\theta}), \tag{2}$$

which maps $\boldsymbol{\theta}$ to a transmission spectrum of fractional transit depths in $D = 52$ wavelength bins. We consider three codes, $c \in \mathcal{C} = \{\mathrm{PSN}, \mathrm{pRT}, \mathrm{TRX}\}$, for POSEIDON [TODO_macdonald_poseidon],

petitRADTRANS [TODO_molliere_petitradtrans], and TauREx [TODO_alrefaie_taurex3]. For a pair of codes (a, b) and the same $\boldsymbol{\theta}$, the inter-code systematic is the difference spectrum

$$\boldsymbol{\Delta}^{(ab)}(\boldsymbol{\theta}) = \mathcal{F}^{(a)}(\boldsymbol{\theta}) - \mathcal{F}^{(b)}(\boldsymbol{\theta}) \in \mathbb{R}^D. \tag{3}$$

In a physically consistent setting with identical inputs, $\boldsymbol{\Delta}^{(ab)} \equiv \mathbf{0}$. We address three questions:

(Q1) *Structure.* What are the amplitude, spectral shape, and parameter dependence of $\boldsymbol{\Delta}^{(ab)}$?

(Q2) *Detectability.* Given a single spectrum, possibly corrupted by observational noise, can one determine which code produced it?

(Q3) *Impact on inference.* How much bias does a data-driven retrieval model trained on code a incur when it is applied to spectra from code $b \neq a$?

We treat (Q1) with a paired residual decomposition (Section 3.3), (Q2) with one-class autoencoders and a supervised classifier (Sections 3.4–3.5), and (Q3) with gradient-boosted regression (Section 3.6).

3.2 Simulated spectra

Parameter sampling. We draw $N = 98{,}725$ parameter vectors [TODO: VERIFY whether 98,725 is the full sample or the sample remaining after a cleaning step, and what that step removed]. In the stored sample, $R_\star$, R_p, M_p, and T are approximately uniform on $[0.5, 2.0]\,R_\odot$, $[0.1, 2.0]\,R_\mathrm{J}$, $[0.01, 5.0]\,M_\mathrm{J}$ [TODO: VERIFY lower bound of M_p; the lowest-mass decile is under-populated by $\approx$12%], and $[300, 3000]\,\mathrm{K}$, and the five log-abundances are uniform on $[-9, -3]$. Two abundances are restricted where opacity data are unavailable at high temperature: $\log_{10} X_{\mathrm{CH_4}} \in [-9, -7]$ for $T > 2000\,\mathrm{K}$ and $\log_{10} X_{\mathrm{NH_3}} \in [-9, -7]$ for $T > 1600\,\mathrm{K}$ [TODO: VERIFY; author notes give 1500 K, but the stored parameters show the cut-offs at 2000 K for CH_4 and 1600 K for NH_3]. This restriction induces negative correlations between T and the two log-abundances (Pearson $r = -0.47$ for CH_4 and -0.53 for NH_3) and a positive correlation between the two abundances themselves ($r = 0.26$); all other parameter pairs are uncorrelated ($|r| < 0.02$).

Forward models. All three codes use an isothermal pressure–temperature profile at T, a cloud-free atmosphere, and each code's default opacity database [TODO: VERIFY line lists, collision-induced absorption, Rayleigh scattering, background gas composition, and reference pressure for each code]. POSEIDON and TauREx compute spectra from cross-sections at a native resolving power of $R = 15{,}000$; petitRADTRANS uses correlated-k opacities at $R = 1000$. Each native spectrum is binned onto a common grid of $D = 52$ bins that follows the Ariel spectral coverage [TODO_tinetti_ariel_mission] [TODO: VERIFY Ariel tier and binning reference]. Bin centres span 0.55–7.28 μm: three photometric bins (0.55, 0.70, 0.95 μm), followed by three spectroscopic segments with approximately constant $\lambda/\Delta\lambda \approx 11$ (1.16–1.88 μm, 6 bins), ≈ 51 (1.97–3.72 μm, 33 bins), and ≈ 16 (4.03–7.28 μm, 10 bins). Because the three codes share $\boldsymbol{\theta}$ row by row, every planet i has a triplet $(\mathbf{y}_i^{(\mathrm{PSN})}, \mathbf{y}_i^{(\mathrm{pRT})}, \mathbf{y}_i^{(\mathrm{TRX})})$, so the effect of the code can be isolated from that of the planet.

Feature-height selection. We define the feature height of a spectrum as

$$h(\mathbf{y}) = \max_j y_j - \min_j y_j, \tag{4}$$

and retain planet i only if $h(\mathbf{y}_i^{(c)}) \geq h_{\min} = 210\,\text{ppm}$ for every code under consideration [TODO: VERIFY physical justification for 210 ppm]. The median feature height of the unselected sample is 153–156 ppm, depending on the code, so the selection retains 44.6% of planets.

Per-spectrum normalisation. For $\mathbf{y} \in \mathbb{R}^D$ we define the bin-averaged depth and the population standard deviation

$$m(\mathbf{y}) = \frac{1}{D}\sum_{j=1}^{D} y_j, \qquad s(\mathbf{y}) = \Big(\frac{1}{D}\sum_{j=1}^{D}\big(y_j - m(\mathbf{y})\big)^2\Big)^{1/2}, \tag{5}$$

and the standardised spectrum $\tilde{\mathbf{y}} = (\mathbf{y} - m(\mathbf{y})\mathbf{1})/s(\mathbf{y})$. Standardisation removes the absolute depth, which encodes $(R_p/R_\star)^2$, and the overall feature amplitude. Models that operate on $\tilde{\mathbf{y}}$ are therefore sensitive only to spectral shape; where absolute depth is required (Section 3.6), m and s are supplied as separate inputs.

Noise model. Observational noise is modelled as white Gaussian noise with a uniform standard deviation σ in every bin,

$$\mathbf{y}^\sigma = \mathbf{y} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D), \tag{6}$$

added to the transit-depth spectrum before standardisation, so that $\tilde{\mathbf{y}}^\sigma$ is computed with $m(\mathbf{y}^\sigma)$ and $s(\mathbf{y}^\sigma)$.

3.3 Paired residual decomposition (Q1)

For each planet i and code pair (a, b) we split $\boldsymbol{\Delta}_i^{(ab)}$ (Eq. 3) into a wavelength-independent offset and a shape residual,

$$\delta_i^{(ab)} = \frac{1}{D}\mathbf{1}^\top \boldsymbol{\Delta}_i^{(ab)}, \qquad \mathbf{r}_i^{(ab)} = \boldsymbol{\Delta}_i^{(ab)} - \delta_i^{(ab)}\mathbf{1}. \tag{7}$$

The offset is degenerate with R_p in a retrieval and is reported separately. We summarise the shape residual by its root-mean-square amplitude,

$$\rho_i^{(ab)} = \frac{\|\mathbf{r}_i^{(ab)}\|_2}{\sqrt{D}}, \tag{8}$$

which is also the noise-free signal available to any detector that operates on mean-subtracted spectra. To compare planets with different feature amplitudes we also form the relative residual $\mathbf{q}_i^{(ab)} = \mathbf{r}_i^{(ab)}/\bar{h}_i^{(ab)}$, with $\bar{h}_i^{(ab)} = \frac{1}{2}[h(\mathbf{y}_i^{(a)}) + h(\mathbf{y}_i^{(b)})]$.

Principal components of the residuals. For each pair we stack the residuals into $\mathbf{Q}^{(ab)} \in \mathbb{R}^{N' \times D}$ (and analogously the absolute residuals), centre the columns, and compute the singular value decomposition $\mathbf{Q}^{(ab)} - \mathbf{1}\bar{\mathbf{q}}^\top = \mathbf{U}\mathbf{S}\mathbf{V}^\top$. The rows $\mathbf{v}_k^\top$ of $\mathbf{V}^\top$ give the spectral shapes of the dominant inter-code differences, each signed so that its largest-magnitude loading is positive. The scores $u_{ik} = (\mathbf{q}_i - \bar{\mathbf{q}})^\top \mathbf{v}_k$ give their per-planet amplitudes, and the explained-variance ratios are $S_{kk}^2 / \sum_{k'} S_{k'k'}^2$.

Parameter dependence. We quantify how deterministically each component depends on the planet parameters by regressing its score on $\boldsymbol{\theta}$ with a random-forest regressor $\mathcal{R}_k : \Theta \to \mathbb{R}$ and reporting the out-of-sample coefficient of determination

$$R_k^2 = 1 - \frac{\sum_{i\in\mathcal{H}} \big(u_{ik} - \mathcal{R}_k(\boldsymbol{\theta}_i)\big)^2}{\sum_{i\in\mathcal{H}} \big(u_{ik} - \bar{u}_k\big)^2}, \tag{9}$$

on a held-out subset $\mathcal{H}$. A value $R_k^2 \to 1$ indicates that component k is a deterministic function of $\boldsymbol{\theta}$ and could, in principle, be emulated and corrected. We additionally report Spearman rank correlations between the scores, the offset, and the parameters together with two derived quantities: the surface gravity proxy $\log_{10} g = \log_{10}(M_p/R_p^2)$ in Jovian units, and $\log_{10}(TR_p/M_p)$, which is proportional to $\log_{10}(H/R_p)$ at fixed mean molecular weight, where H is the atmospheric scale height.

3.4 One-class detection with autoencoders (Q2)

Architecture. For each code c we train a fully connected autoencoder $f_c = g_\psi \circ e_\phi$ on standardised spectra from that code only. Let $(d_1, \dots, d_K)$ be the hidden widths and L the latent dimension. With $\mathbf{a}_0 = \tilde{\mathbf{y}}$, the encoder computes

$$\mathbf{a}_\ell = \varphi(\mathbf{W}_\ell \mathbf{a}_{\ell-1} + \mathbf{b}_\ell), \qquad \ell = 1, \dots, K, \tag{10}$$

$$\mathbf{z} = \mathbf{W}_{K+1}\mathbf{a}_K + \mathbf{b}_{K+1} \in \mathbb{R}^L, \tag{11}$$

with $\mathbf{W}_\ell \in \mathbb{R}^{d_\ell \times d_{\ell-1}}$, $d_0 = D$, and $d_{K+1} = L$. The decoder mirrors the encoder: with $\mathbf{a}'_0 = \mathbf{z}$ and widths $(d_K, \dots, d_1)$,

$$\mathbf{a}'_\ell = \varphi(\mathbf{W}'_\ell \mathbf{a}'_{\ell-1} + \mathbf{b}'_\ell), \qquad \ell = 1, \dots, K, \tag{12}$$

$$\hat{\tilde{\mathbf{y}}} = \mathbf{W}'_{K+1}\mathbf{a}'_K + \mathbf{b}'_{K+1} \in \mathbb{R}^D. \tag{13}$$

The latent and output layers are linear because standardised spectra are unbounded. The encoder and decoder do not share weights, and the network has no normalisation, dropout, or skip connections. Unless stated otherwise, φ is the Gaussian error linear unit $\varphi(x) = x\,\Phi(x)$ [TODO_hendrycks_gelu], with Φ the standard normal cumulative distribution function. We use two widths:

$$\text{AE-S: } (d_1, \dots, d_4) = (128, 64, 32, 16), \qquad \text{AE-W: } (d_1, \dots, d_5) = (256, 128, 64, 32, 16), \tag{14}$$

both with $L = 8$. The number of trainable parameters is

$$P = 2\sum_{\ell=1}^{K+1} d_{\ell-1}d_\ell + \sum_{\ell=1}^{K+1} d_\ell + \sum_{\ell=0}^{K} d_\ell, \tag{15}$$

which gives $P = 35{,}612$ for AE-S and $P = 114{,}972$ for AE-W (Table 1). AE-W adds one 256-unit layer at each end of AE-S and has 3.2× as many parameters.

Reconstruction objective. Given a mini-batch $\mathcal{B}$, the autoencoder minimises the mean squared reconstruction error in standardised units,

$$\mathcal{L}_{\text{rec}}(\boldsymbol{\phi}, \boldsymbol{\psi}) = \frac{1}{|\mathcal{B}|\,D} \sum_{i\in\mathcal{B}} \big\|\tilde{\mathbf{y}}_i - f_c(\tilde{\mathbf{y}}_i)\big\|_2^2. \tag{16}$$

Table 1: Autoencoder variants. All variants are symmetric, use linear latent and output layers, and differ only in the hidden widths, the latent dimension L, and the activation φ. MACs: multiply–accumulate operations per forward pass.

Variant	Encoder widths	L	Parameters	MACs
AE-S	$52 \to 128 \to 64 \to 32 \to 16$	8	35,612	35,072
AE-S(L)	as AE-S	4–16	$35{,}348 + 33L$	–
AE-W	$52 \to 256 \to 128 \to 64 \to 32 \to 16$	8	114,972	113,920

Parameters $\boldsymbol{\omega} = (\boldsymbol{\phi}, \boldsymbol{\psi})$ are updated with AdamW [TODO_loshchilov_adamw], which applies decoupled weight decay λ,

$$\boldsymbol{\omega}_{t+1} = (1 - \eta_t \lambda)\, \boldsymbol{\omega}_t - \eta_t \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon}, \tag{17}$$

where $\hat{\mathbf{m}}_t$ and $\hat{\mathbf{v}}_t$ are the bias-corrected first and second moment estimates of $\nabla_{\boldsymbol{\omega}} \mathcal{L}_{\mathrm{rec}}$.

Denoising objective. To test whether robustness to noise can be learned, we also train denoising autoencoders [TODO_vincent_denoising_ae] with the AE-W architecture. The input is the standardised noisy spectrum. The target is the clean spectrum scaled by the *noisy* spectrum's statistics, so that the output can be mapped back to transit depth from observable quantities only:

$$\mathcal{L}_{\mathrm{den}} = \frac{1}{|\mathcal{B}|\, D} \sum_{i \in \mathcal{B}} \left\| \frac{\mathbf{y}_i - m(\mathbf{y}_i^{\sigma})\mathbf{1}}{s(\mathbf{y}_i^{\sigma})} - f_c(\tilde{\mathbf{y}}_i^{\sigma}) \right\|_2^2. \tag{18}$$

A new noise realisation $\boldsymbol{\epsilon}$ is drawn for every training mini-batch.

Anomaly scores. An autoencoder trained on code c assigns a spectrum $\mathbf{y}$ the reconstruction score

$$e_c(\mathbf{y}) = \frac{1}{D} \| \tilde{\mathbf{y}} - f_c(\tilde{\mathbf{y}}) \|_2^2. \tag{19}$$

Spectra from code c are treated as nominal and spectra from $b \neq c$ as anomalous. Decision thresholds $\tau_c^{(q)}$ are the q-th percentiles, $q \in \{95, 99\}$, of e_c on the nominal validation spectra, so no anomalous data are used to set them.

For noisy spectra with known per-bin uncertainty σ, e_c conflates reconstruction error with noise. We therefore also map the reconstruction back to transit depth and score it against the error bars:

$$\hat{\mathbf{y}} = s(\mathbf{y}^{\sigma})\, f_c(\tilde{\mathbf{y}}^{\sigma}) + m(\mathbf{y}^{\sigma})\mathbf{1}, \qquad \chi^2_{\nu,c}(\mathbf{y}^{\sigma}) = \frac{1}{D} \sum_{j=1}^{D} \left(\frac{y_j^{\sigma} - \hat{y}_j}{\sigma} \right)^2. \tag{20}$$

We compare two thresholds at a target false-positive rate α. The first is the theoretical threshold $\tau^{\mathrm{th}} = F^{-1}_{\chi^2_D}(1 - \alpha)/D$, where $F_{\chi^2_D}$ is the χ^2 cumulative distribution function with D degrees of freedom. The second is an empirical threshold τ_c^{emp}, defined as the $(1-\alpha)$ quantile of $\chi^2_{\nu,c}$ on noisy nominal validation spectra, which absorbs the autoencoder's own reconstruction error.

Oracle reference. To separate the limits imposed by the data from those of the model, we define an oracle that scores a noisy spectrum against the true noise-free spectrum of code c for the same planet:

$$\chi^{2\,\star}_{\nu,c}(\mathbf{y}_i^{\sigma}) = \frac{1}{D} \sum_{j=1}^{D} \left(\frac{y_{ij}^{\sigma} - y_{ij}^{(c)}}{\sigma} \right)^2. \tag{21}$$

The oracle requires the true $\boldsymbol{\theta}_i$ and is not realisable in practice. It bounds the performance of any single-spectrum goodness-of-fit test against code c.

3.5 Supervised code classification (Q2)

As a discriminative counterpart to the one-class detectors, we train a three-class random forest [TODO_breiman_random_forests] $\mathcal{G} : \mathbb{R}^D \to \Delta^2$ that maps a standardised (noisy) spectrum to class probabilities $\mathbf{p} = (p_{\mathrm{PSN}}, p_{\mathrm{pRT}}, p_{\mathrm{TRX}})$ over the three codes. Each training planet contributes one spectrum per code, so the classifier sees every planet under every label and must rely on code-specific differences rather than planet properties. To compare with the one-class detectors, for a pair (a, b) we score test spectra of codes a and b with

$$\pi_{ab}(\mathbf{y}) = \frac{p_b(\mathbf{y})}{p_a(\mathbf{y}) + p_b(\mathbf{y})}, \tag{22}$$

which plays the role of an anomaly score for nominal code a.

3.6 Self- and cross-code retrieval (Q3)

We emulate a data-driven retrieval with gradient-boosted regression trees [TODO_chen_xgboost]. The input feature vector retains the absolute depth information removed by standardisation,

$$\mathbf{x}(\mathbf{y}) = \left[\, \tilde{\mathbf{y}}^{\top},\ \log_{10} m(\mathbf{y}),\ \log_{10} s(\mathbf{y}) \,\right]^{\top} \in \mathbb{R}^{D+2}. \tag{23}$$

For each training code a and each parameter θ_k, $k = 1, \ldots, 9$, we fit an independent regressor $\hat{\theta}_k^{(a)} = G_k^{(a)}(\mathbf{x}(\mathbf{y}^{(a)}))$, which gives 27 regressors in total. Each regressor is an additive ensemble of regression trees fitted by second-order boosting on the squared error. Every regressor is then applied to the test spectra of all three codes for the same planets. We call the case $b = a$ *self-retrieval* and the case $b \neq a$ *cross-retrieval*. Because the planets are identical, any difference between self- and cross-retrieval errors is caused solely by $\boldsymbol{\Delta}^{(ab)}$.

For each triple (a, b, k) with errors $\varepsilon_i = \hat{\theta}_k^{(a)}(\mathbf{y}_i^{(b)}) - \theta_{ik}$ on the test set, we report the coefficient of determination R^2 (Eq. 9), the median bias $\beta = \mathrm{median}_i\, \varepsilon_i$, and the half-width

$$w = \tfrac{1}{2}\left[P_{84}(\varepsilon) - P_{16}(\varepsilon)\right], \tag{24}$$

where P_q is the q-th percentile. We also report β as a percentage of the parameter's range over the sample. The ratio $|\beta|/w$ measures how large the code-induced bias is relative to the single-planet scatter.

3.7 Diagnostics of reconstruction difficulty

To identify which planets limit the autoencoders, we rank each code's held-out spectra by e_c and label the top 5% as poorly reconstructed. For every parameter and derived quantity ξ, we compute the probability that a poorly reconstructed planet has a larger value of ξ than another planet,

$$A_\xi = \Pr(\xi_i > \xi_{i'} \mid i \in \mathcal{P},\ i' \notin \mathcal{P}), \tag{25}$$

where $\mathcal{P}$ is the poorly reconstructed set. $A_\xi = 0.5$ indicates no association. We measure agreement across codes by the overlap of the sets $\mathcal{P}$ and by the Spearman correlation of per-planet scores. An analogous breakdown of classifier accuracy is computed in quantile bins of $\boldsymbol{\theta}$, h, and the mean pairwise residual amplitude $\bar{\rho}_i = \frac{1}{3}\sum_{(a,b)} \rho_i^{(ab)}$.

Table 2: Autoencoder training hyperparameters. Values marked † are framework defaults.

Hyperparameter	Value
Loss	$\mathcal{L}_{\text{rec}}$ (Eq. 16) or $\mathcal{L}_{\text{den}}$ (Eq. 18)
Optimiser	AdamW, $\eta_0 = 10^{-3}$, $\lambda = 10^{-5}$
$(\beta_1, \beta_2, \epsilon)$	(0.9, 0.999, 10^{-8})† [TODO: VERIFY framework version]
Mini-batch size	256
Maximum epochs	1000
Plateau schedule	factor 0.5, patience 7 epochs, no minimum η
Cosine schedule	$\eta_{\min} = 10^{-6}$, $E = 1000$
Early stopping	$\delta = 10^{-3}$ (relative); $P_{\text{es}} = 20$ (plateau) or 50 (cosine)
Activation φ	GELU (exact form)
Latent dimension L	8

4 Experimental Setup

4.1 Data partitions

We use two planet sets. For the autoencoder design study (Section 4.3), the feature-height selection is applied to POSEIDON spectra only, which retains 44,026 planets. For every cross-code experiment, a planet must pass the selection in all three codes, which retains 43,885 planets. In both cases a single random permutation (seed 42) assigns 70%/15%/15% of planets to training, validation, and test sets (30,818/6,604/6,604 and 30,719/6,583/6,583, respectively). In the cross-code set the partition is defined on planets, so a test planet is absent from training in every code. The validation set is used for early stopping, model selection, and threshold calibration. The test set is used only for the reported metrics.

4.2 Autoencoder training protocol

Table 2 lists the training hyperparameters. We train for at most 1000 epochs with mini-batches of 256 spectra and use one of two learning-rate schedules. The *plateau* schedule halves η when the validation loss has not improved for 7 epochs, with no lower bound on η. The *cosine* schedule anneals η once per epoch t as

$$\eta_t = \eta_{\min} + \tfrac{1}{2}(\eta_0 - \eta_{\min})(1 + \cos(\pi t/E)), \tag{26}$$

with $\eta_0 = 10^{-3}$, $\eta_{\min} = 10^{-6}$, and $E = 1000$. Early stopping tracks a reference loss $\mathcal{L}^{\text{ref}}$ and resets its counter when the validation loss satisfies $\mathcal{L}_{\text{val}} < (1 - \delta)\,\mathcal{L}^{\text{ref}}$, with relative tolerance $\delta = 10^{-3}$. Training stops after P_{es} consecutive epochs without such an improvement. We use $P_{\text{es}} = 20$ with the plateau schedule and $P_{\text{es}} = 50$ with the cosine schedule. The weights with the lowest validation loss are restored at the end of training. Weights and biases use the default initialisation, drawn from $\mathcal{U}(-d_{\ell-1}^{-1/2}, d_{\ell-1}^{-1/2})$. The initialisation seed also determines the mini-batch order.

4.3 Autoencoder design study

The autoencoders in this work serve as measuring instruments. A detector whose own reconstruction error exceeds the inter-code signal cannot measure that signal. We therefore select a configuration that minimises the reconstruction error before using it for detection. All design choices use

the POSEIDON-only planet set and are made on validation MSE. Selecting on test MSE would give the same configuration at every stage.

1. *Linear baseline.* Principal component analysis fitted on the training spectra, with $k \in \{4, \ldots, 16\}$ components, provides the reconstruction error of the optimal linear encoder of each size.

2. *Latent dimension.* AE-S with $L \in \{4, \ldots, 16\}$, plateau schedule, one seed (42). We select the smallest L at which validation MSE reaches its plateau.

3. *Activation function.* AE-S with $L = 8$ and $\varphi \in$ {ReLU, leaky ReLU, ELU, SELU, GELU, SiLU, Mish, tanh, softplus}, plateau schedule, three seeds (42, 43, 44).

4. *Width and schedule.* A 2×2 design over {AE-S, AE-W} $\times$ {plateau with $P_{\mathrm{es}} = 20$, cosine with $P_{\mathrm{es}} = 50$}, seeds 42 and 43, with a third seed (44) for AE-W with the cosine schedule. The schedule and the early-stopping patience vary jointly in this design, so their individual contributions are not separated.

The selected configuration, AE-W with GELU, $L = 8$, and the cosine schedule, is used for all cross-code experiments.

4.4 Detection experiments

Noise-free detection. One AE-W model is trained per code on the cross-code planet set. For POSEIDON and petitRADTRANS we train three initialisation seeds (42, 43, 44); for TauREx, one seed (42) [TODO: VERIFY whether to equalise seeds before submission]. For each code, the run with the lowest own-code validation loss is used. We report the area under the receiver operating characteristic curve (AUC) for each of the six ordered pairs (c, b), computed with the Mann–Whitney statistic

$$\mathrm{AUC}_{cb} = \frac{1}{n_0 n_1} \sum_{i=1}^{n_0} \sum_{i'=1}^{n_1} \mathbb{1}\big[e_c(\mathbf{y}_{i'}^{(b)}) > e_c(\mathbf{y}_i^{(c)})\big], \tag{27}$$

together with the fractions of test spectra above $\tau_c^{(95)}$ and $\tau_c^{(99)}$.

Detection under noise. The noise-free models are applied without retraining to test spectra corrupted according to Eq. (6) with $\sigma \in \{2, 5, 10, 20, 30, 50, 100\}$ ppm. AUCs are averaged over two independent noise realisations per σ. For the χ^2 score we use $\alpha = 0.01$, which gives $\tau^{\mathrm{th}} = 1.512$. We estimate this value from 2×10^6 samples of χ^2_{52}. The empirical threshold τ_c^{emp} is computed on validation spectra with independent noise realisations at the same σ. We report the true-positive rate at each threshold, the realised false-positive rate on nominal test spectra, and the oracle of Eq. (21).

Denoising models. We train 12 denoising AE-W models, one for each code and each $\sigma \in \{10, 20, 30, 50\}$ ppm, with seed 42 and otherwise the same protocol. Validation and test spectra use one fixed noise realisation each. These models are evaluated with the same χ^2 protocol at the noise level used in training.

Random-forest classifier. The classifier uses 300 trees, at least 2 samples per leaf, and $\lfloor\sqrt{D}\rfloor = 7$ candidate features per split, with Gini impurity and bootstrap sampling as defaults. For each $\sigma \in \{0, 10, 20, 30, 50\}$ ppm we add one independent noise realisation to every training and test spectrum, standardise, and train and test at the same σ. This gives $3 \times 30{,}719 = 92{,}157$ training spectra and $3 \times 6{,}583 = 19{,}749$ test spectra. We report accuracy, per-class recall, macro-averaged one-versus-rest AUC, and the mean pairwise AUC under the score in Eq. (22).

4.5 Retrieval experiments

The XGBoost [TODO_chen_xgboost] regressors use learning rate 0.1, maximum depth 8, row and column subsampling 0.8, minimum child weight 2, and histogram-based split finding. Each is trained for up to 3000 boosting rounds, with early stopping after 100 rounds without improvement in validation RMSE. Models are trained on noise-free spectra. To measure how the self/cross gap scales with data volume, we retrain on random subsets containing 5%, 10%, 25%, and 50% of the training planets. To measure the interaction between code systematics and noise, the noise-free models are applied without retraining to test spectra with $\sigma \in \{10, 20, 30, 50\}$ ppm. Because these models never see noise during training, their noisy-test results are a lower bound on what a noise-aware retrieval would achieve.

4.6 Residual analysis settings

The paired decomposition uses all 43,885 planets of the cross-code set without noise. For each of the three unordered pairs we compute the first four principal components of the absolute and relative residuals. We regress each score on $\boldsymbol{\theta}$ with a random forest of 100 trees, at least 5 samples per leaf, and half of the features per split, and report R^2 on a random 30% held-out subset.

4.7 Implementation

Neural models are implemented in PyTorch [TODO_paszke_pytorch], the random forests and metrics in scikit-learn [TODO_pedregosa_sklearn], and the boosted trees in XGBoost [TODO_chen_xgboost] [TODO: VERIFY library versions]. All models are trained on CPU [TODO: VERIFY processor model and memory], with each autoencoder run restricted to a single thread and three to four runs executed in parallel. One 1000-epoch run takes about 650 s for AE-S and 920–1130 s for AE-W. Code and trained models are available at [TODO: VERIFY repository URL / archive DOI].

5 Results and Analysis

We first establish that the autoencoder reconstructs spectra accurately enough to serve as a detector (Section 5.1). We then report the structure of the inter-code systematics (Q1, Section 5.2), their detectability with and without noise (Q2, Section 5.3), and their effect on retrieval (Q3, Section 5.4). Section 5.5 compares computational cost. Unless stated otherwise, reconstruction errors are mean squared errors in standardised units on the test set, and uncertainties are sample standard deviations over initialisation seeds.

5.1 Autoencoder design study

Linear versus nonlinear compression. Table 3 compares principal component analysis (PCA) with the four autoencoder configurations of the 2×2 width–schedule design. With the same eight-

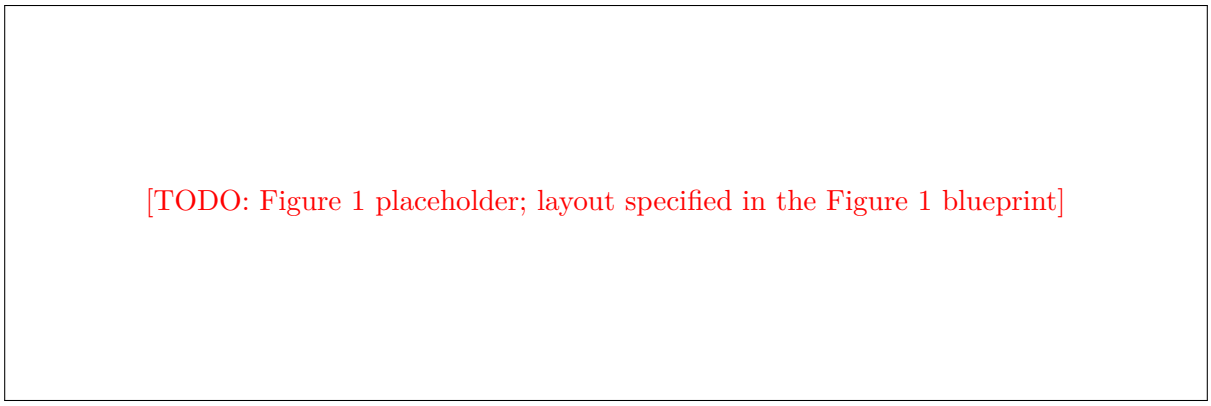


Figure 1: Overview of the study. (a) Each parameter vector $\boldsymbol{\theta}$ is passed through three radiative-transfer codes and binned to the same 52 Ariel bins, giving one spectrum per code for every planet. (b) Preprocessing: feature-height selection, optional white noise, and per-spectrum standardisation. (c) Q1: paired residuals are split into an offset and a shape term, decomposed by PCA, and regressed on $\boldsymbol{\theta}$. (d) Q2: one-class autoencoders, one per code, score spectra by reconstruction error or χ^2_ν and are compared with a three-class random forest and a noise-limited oracle. (e) Q3: XGBoost regressors trained on one code retrieve $\boldsymbol{\theta}$ from all three codes, giving a 3×3 self/cross matrix.

dimensional code, every autoencoder reconstructs at least 58× more accurately than PCA. AE-W with the cosine schedule reaches a test MSE of 6.55×10^{-5}, about 200× below PCA with $k = 8$ components (1.30×10^{-2}). It is also 24× below PCA with $k = 16$ (1.59×10^{-3}), which uses twice as many latent variables. PCA needs 13, 25, and 39 components to explain 99%, 99.9%, and 99.99% of the variance. The standardised spectra therefore lie on a low-dimensional but curved manifold. A linear basis must spend many components to approximate this curvature; a nonlinear encoder can parameterise it directly. This agrees with the forward model of Eq. (2): transit depth depends on $\boldsymbol{\theta}$ through exponential and logarithmic terms in the optical depth and scale height [TODO_benneke_seager_transmission_scaleheight], and these terms do not combine linearly.

Ablation: width and schedule. The cosine schedule with $P_{\rm es} = 50$ lowers the AE-S test MSE from 2.08×10^{-4} to 1.10×10^{-4} (1.9×). It also reduces the spread across seeds from $\pm 1.7 \times 10^{-5}$ to $\pm 4 \times 10^{-6}$. Both cosine runs train for the full 1000 epochs. For the seed that completed training under both widths (43), widening the network lowers the test MSE from 2.20×10^{-4} to 1.20×10^{-4} (1.8×). Combining both changes lowers it to 6.55×10^{-5} (3.2×), so the two effects approximately multiply. The plateau schedule shows a failure mode: for AE-W with seed 42, repeated halving drove the learning rate to 3.9×10^{-6} by epoch 334. The run then met the early-stopping criterion at a test MSE of 3.23×10^{-4}, 2.7× worse than the same architecture with seed 43. The schedule and the patience change together in this design, so we attribute the improvement to the pair, not to the schedule alone. For AE-W with the cosine schedule, the training loss at the selected epoch (3.9–4.4×10^{-5} for seeds 42 and 43) falls below the validation loss (5.7–5.9×10^{-5}). This train–validation gap is absent for AE-S. AE-W is therefore no longer limited by capacity, and further gains would require regularisation or more data rather than width.

Ablation: latent dimension. Table 4 reports the bottleneck sweep. The test MSE falls by a factor of 17 from $L = 4$ to $L = 8$ and is flat within single-seed variation for $L \geq 8$ (1.6–2.4×10^{-4}).

Table 3: Reconstruction of POSEIDON spectra by linear and nonlinear compressors with an 8-dimensional code (PCA is also shown with $k = 16$). Errors are MSE in standardised units ($\times 10^{-4}$). Autoencoder rows are means over the two seeds shared by all four configurations (42, 43); the test column also gives the seed standard deviation. With a third seed, AE-W / cosine gives 0.80 ± 0.26. Bold: best; underline: second best. PCA parameters count the basis vectors and the mean; MACs count encoding plus decoding.

Model	Schedule / $P_{\rm es}$	Params	MACs	Time (s)	Val	Test	Median	P_{95}
PCA, $k = 8$	–	468	832	–	–	130	84.9	385
PCA, $k = 16$	–	884	1,664	–	–	15.9	10.4	44.6
AE-S	plateau / 20	35,612	35,072	411	1.98	2.08 ± 0.17	0.95	6.33
AE-S	cosine / 50	35,612	35,072	651	<u>1.05</u>	<u>1.10 ± 0.04</u>	<u>0.49</u>	<u>3.40</u>
AE-W	plateau / 20	114,972	113,920	662	2.10	2.22 ± 1.44	0.64	7.51
AE-W	cosine / 50	114,972	113,920	1123	**0.58**	$\mathbf{0.66 \pm 0.08}$	**0.20**	**1.85**

Two exceptions, $L = 7$ and $L = 12$, stopped early, after 359 and 356 epochs, on a transient plateau of the validation loss. The lowest validation MSE occurs at $L = 9$, but the difference from $L = 8$ (1.64 versus 1.98×10^{-4}) is smaller than the seed-to-seed spread at fixed L ($\pm 1.1 \times 10^{-4}$; Table 5). We therefore adopt $L = 8$ as the smallest dimension on the plateau. The spectra are generated from nine parameters, which bounds the intrinsic dimension of the standardised spectra from above. Standardisation removes the absolute depth and the amplitude, two functions of $\boldsymbol{\theta}$, so the shape manifold can have fewer than nine dimensions.

Ablation: activation function. Table 5 compares nine activations with AE-S, $L = 8$, and the plateau schedule, over three seeds. GELU has the lowest mean validation and test MSE. ReLU has the highest test MSE, 8.44×10^{-4}, which is 3.1× that of GELU. Spectra are smooth functions of wavelength and of $\boldsymbol{\theta}$, whereas a ReLU network is piecewise linear in its input. It must therefore approximate a curved manifold with a finite number of linear pieces, which is a weaker inductive bias for this data than a smooth activation. Smoothness is not sufficient on its own: softplus is smooth but ranks second to last. GELU, SiLU, Mish, tanh, and ELU each reach 1.8–2.6×10^{-4} on their best seed and overlap within seed variation; no seed of ReLU, leaky ReLU, SELU, or softplus goes below 3.9×10^{-4}. Within this group, run length explains most of the variation: the Spearman correlation between the number of epochs before early stopping and the test MSE is -0.85 ($n = 15$). The six runs that stopped before 450 epochs end at 4.0–5.8×10^{-4}, and the four that ran beyond 750 epochs reach 1.8–2.4×10^{-4}. The ranking among smooth activations is therefore dominated by when the plateau schedule stops training, consistent with the schedule ablation above.

Autoencoder error in physical units. For the cross-code models (AE-W, cosine schedule), the own-code test MSE is 1.96×10^{-4} for POSEIDON, 1.56×10^{-4} for petitRADTRANS, and 5.54×10^{-5} for TauREx. Multiplying the per-spectrum root-mean-square error by $s(\mathbf{y})$ gives the reconstruction error in transit depth. Its median is 0.9–1.2 ppm across the three codes, with 99th percentiles of 85–135 ppm. For 98.7–99.9% of test planets the own-code reconstruction error is smaller than the inter-code residual amplitude $\rho_i^{(ab)}$ of the same planet. For typical planets, the autoencoders therefore resolve differences well below the inter-code signal. The heavy upper tail of the error distribution limits detection, as Section 5.3 shows.

Table 4: Bottleneck sweep for AE-S with the plateau schedule (one seed). MSE in units of 10^{-4}. $L = 7$ and $L = 12$ stopped early on a transient plateau.

L	Val	Test	Epochs
4	35.4	33.7	320
5	13.2	13.1	358
6	4.32	4.37	395
7	5.16	5.25	359
8	1.98	1.96	747
9	**1.64**	**1.64**	648
10	1.95	1.97	726
11	2.15	2.12	841
12	4.91	4.90	356
13	1.85	1.83	1000
14	2.48	2.41	661
15	1.85	1.80	579
16	<u>1.77</u>	<u>1.78</u>	720

Table 5: Activation ablation (AE-S, $L = 8$, plateau schedule, three seeds). MSE in units of 10^{-4}; the median column averages the per-run medians. Bold: best; underline: second best.

φ	Val	Test	Best seed	Median	Epochs
ReLU	8.08	8.44 ± 2.98	6.56	4.56	579
Leaky ReLU	4.54	4.60 ± 0.32	4.31	2.43	511
Softplus	5.72	5.87 ± 0.24	5.67	2.01	409
SELU	4.27	4.34 ± 0.36	3.93	2.31	406
ELU	3.71	3.80 ± 1.74	2.59	1.57	420
Mish	3.55	3.69 ± 1.63	**1.83**	1.26	586
SiLU	3.53	3.58 ± 1.38	2.01	<u>1.25</u>	601
tanh	<u>3.23</u>	<u>3.38 ± 1.77</u>	2.34	1.34	715
GELU	**2.75**	$\mathbf{2.73 \pm 1.13}$	<u>1.96</u>	**1.12**	571

5.2 Q1: Structure of the inter-code systematics

Table 6 summarises the paired decomposition of Eq. (7) over the 43,885 planets of the cross-code set.

Offsets. POSEIDON spectra are systematically shallower than those of the other two codes, with median offsets of -24 and -21 ppm. The offset magnitude scales with feature height: its Spearman correlation with h is -0.96 (PSN$-$pRT) and -0.89 (PSN$-$TRX). It also correlates with $\log_{10}(H/R_p)$, with $\rho_S = -0.64$ and -0.59. A difference that grows with the scale height and the feature amplitude is consistent with a different reference radius or reference pressure, or a different treatment of the scale height, in POSEIDON [TODO: VERIFY reference pressure and radius conventions of each code]. petitRADTRANS and TauREx agree on the offset to within 2 ppm in the median. The offset is degenerate with R_p and is removed by standardisation, so it is invisible to the detectors of Section 5.3. It nevertheless propagates into retrieved radii (Section 5.4).

Table 6: Paired inter-code residuals on noise-free spectra. δ: wavelength-mean offset; ρ: root-mean-square shape residual after offset removal (Eq. 8); $\rho/\bar{h}$: median relative residual. n_{90}, n_{99}: number of principal components of the relative residual needed for 90% and 99% of its variance. $R^2_{1\text{–}4}$: out-of-sample R^2 of the first four component scores regressed on $\boldsymbol{\theta}$, for relative and absolute residuals.

Pair ($a-b$)	δ median [P_{16}, P_{84}] (ppm)	ρ median (ppm)	ρ P_{90} (ppm)	$\rho/\bar{h}$	n_{90}/n_{99}	$R^2_{1\text{–}4}$ (relative)	$R^2_{1\text{–}4}$ (absolute)
PSN − pRT	−24.4 [−109.3, −9.4]	10.2	91.2	1.3%	6/22	0.99, 0.96, 0.96, 0.95	0.40–0.62
PSN − TRX	−20.8 [−85.8, −7.8]	10.0	75.0	1.3%	7/24	0.98, 0.94, 0.96, 0.95	0.28–0.65
pRT − TRX	+2.1 [−1.1, +20.7]	8.3	74.8	1.0%	6/21	0.99, 0.96, 0.94, 0.96	0.10–0.54

Shape residuals. After offset removal, the codes agree to a median of 8–10 ppm, or 1.0–1.3% of the feature height. The 90th percentile is 75–91 ppm; these are the planets with the largest features. The residuals are not white. The relative residuals require only 6–7 components to capture 90% of their variance, and each leading component has a well-defined spectral location:

- Both pairs that involve petitRADTRANS are dominated by a component (55–60% of the variance) concentrated in three narrow bins at 1.55, 2.31, and 2.36 μm. Its amplitude tracks T and $\log_{10} X_{CO_2}$ and anti-tracks $\log_{10} X_{H_2O}$ ($|\rho_S| = 0.39$–0.52; the sign of each correlation depends on the order of the pair). Localised differences of this kind are expected where correlated-k opacities at $R = 1000$ and cross-sections at $R = 15{,}000$ sample strong, narrow band structure differently [TODO_correlated_k_vs_line_by_line_accuracy]. We list this as a hypothesis; a band-by-band comparison of opacities is required to confirm it.
- The leading component of PSN−TRX (60%) is a broad deficit at short wavelengths, 0.55–1.27 μm, driven almost entirely by $\log_{10} X_{H_2O}$ ($\rho_S = -0.78$). This pattern indicates a difference in the short-wavelength continuum (Rayleigh scattering or collision-induced absorption) or in the near-infrared water opacity [TODO: VERIFY by comparing the default continuum opacities of the two codes].

Determinism of the systematics. Each of the first four relative components is predicted from $\boldsymbol{\theta}$ with out-of-sample $R^2 \geq 0.94$. The absolute components are less predictable ($R^2 = 0.10$–0.65), because their amplitude also scales with the feature height, which the regression does not receive directly. To first order, the inter-code systematics are therefore a deterministic, low-dimensional function of the planet parameters. They could be emulated and either corrected or included as a modelled error term in retrievals.

5.3 Q2: Detectability of the generating code

Noise-free detection. Table 7 lists the one-class detection performance on noise-free spectra. Every ordered pair is separated with AUC ≥ 0.962. Spectra from another code have median reconstruction errors 42–74× higher than the training code's own spectra. At the 95th-percentile threshold the detectors flag 78–98% of other-code spectra, and 4.8–5.0% of own-code spectra, which is the nominal rate. The TauREx model separates best (AUC = 0.992 and 0.988). It also has the lowest own-code reconstruction error, which produces a tighter nominal distribution and a lower threshold. The detectors remain effective for planets with small inter-code differences. For planets with $\rho^{(ab)} < 5$ ppm, which make up 26–33% of the test set, the AUC is still 0.934–0.993, against 0.960–0.997 for larger $\rho^{(ab)}$. Without noise, the inter-code signal is therefore resolved even at the few-ppm level.

Table 7: One-class detection on noise-free test spectra (6,583 planets per code). The ratio column gives the median error on other-code spectra divided by the median error on own-code spectra. Flagged fractions use the validation-set thresholds $\tau_c^{(95)}$ and $\tau_c^{(99)}$. Bold: best; underline: second best.

Trained on	Tested on	AUC	Ratio	Flag $\tau^{(95)}$	Flag $\tau^{(99)}$
PSN	pRT	0.962	42	83.2%	29.4%
PSN	TRX	0.964	46	78.2%	33.5%
pRT	PSN	0.970	56	87.7%	38.3%
pRT	TRX	0.968	49	84.9%	32.6%
TRX	PSN	**0.992**	**74**	**97.7%**	**79.5%**
TRX	pRT	<u>0.988</u>	<u>58</u>	<u>95.8%</u>	<u>78.4%</u>

Table 8: Code separability versus white noise: AUC averaged over the six ordered code pairs. e_c: standardised reconstruction error (Eq. 19); χ^2_ν: error-bar-weighted score (Eq. 20); RF: pairwise score of the supervised classifier (Eq. 22), trained and tested at the same σ. The oracle (Eq. 21) requires the true parameters and is not ranked. Bold: best; underline: second best among realisable methods. –: not evaluated.

Method	$\sigma = 0$	2	5	10	20	30	50	100 ppm
AE, e_c	<u>0.974</u>	**0.909**	<u>0.746</u>	0.628	0.557	0.534	0.517	<u>0.506</u>
AE, χ^2_ν	–	<u>0.834</u>	**0.794**	<u>0.735</u>	<u>0.669</u>	<u>0.632</u>	<u>0.596</u>	**0.561**
Denoising AE, χ^2_ν	–	–	–	0.680	0.630	0.597	0.559	–
Random forest	**0.999**	–	–	**0.945**	**0.886**	**0.838**	**0.775**	–
Oracle $\chi^{2\star}_\nu$	–	*0.999*	*0.985*	*0.949*	*0.876*	*0.821*	*0.748*	*0.662*

Ablation: model selection over seeds. With a single initialisation seed, the POSEIDON and petitRADTRANS detectors reach AUC = 0.945–0.956. Selecting the run with the lowest own-code validation loss among three seeds raises this to 0.962–0.970. The selection lowers the own-code validation MSE from 2.28 to 2.00×10^{-4} (POSEIDON) and from 2.56 to 1.51×10^{-4} (petitRADTRANS). Detection performance is therefore set by the quality of the nominal reconstruction, not by any property of the anomalous data, which are never used for selection. The TauREx detector uses a single seed and still has the best reconstruction. Its advantage is therefore not an artefact of selecting over more seeds.

Detection under noise. Table 8 reports the mean AUC over the six code pairs as a function of σ. The reconstruction score e_c degrades fastest: from 0.974 without noise to 0.746 at 5 ppm and 0.628 at 10 ppm, approaching chance by 30 ppm. Two factors explain this. First, the median inter-code residual is $\rho \approx 9$–11 ppm, so at $\sigma \gtrsim 10$ ppm the noise power per bin exceeds the signal for most planets. Second, e_c is computed in standardised units and has no notion of the measurement uncertainty: it penalises noise and inter-code differences equally. The χ^2_ν score of Eq. (20) weights residuals by the error bars. It outperforms e_c at every $\sigma \geq 5$ ppm and roughly doubles the margin above chance at 10–50 ppm (0.735 versus 0.628 at 10 ppm; 0.596 versus 0.517 at 50 ppm). At $\sigma = 2$ ppm, e_c performs better, because the autoencoder's own error tail is large compared with the error bars.

Table 9: Detection at a nominal 1% false-positive rate, mean over code pairs. FPR: own-code test spectra flagged. TPR: other-code test spectra flagged. "Theory": $\tau^{\rm th} = 1.512$; "Empirical": 99th percentile of χ^2_ν on noisy own-code validation spectra. Bold: best; underline: second best among realisable rows, for TPR at the empirical threshold.

Model	Metric	10	20	30	50 ppm
AE	FPR (theory)	14.0%	9.3%	7.5%	6.6%
	TPR (theory)	49.2%	31.8%	24.3%	18.0%
	FPR (empirical)	0.8%	0.8%	0.8%	0.8%
	TPR (empirical)	**4.1**%	**4.1**%	**4.1**%	**4.2**%
Denoising AE	FPR (theory)	22.4%	14.7%	12.0%	10.5%
	TPR (theory)	49.3%	32.4%	24.5%	17.7%
	FPR (empirical)	0.9%	0.8%	0.8%	0.8%
	TPR (empirical)	<u>2.5%</u>	<u>2.5%</u>	<u>2.1%</u>	<u>1.6%</u>
Oracle	*FPR (theory)*	*1.0%*	*1.0%*	*1.0%*	*0.9%*
	TPR (theory)	*81.3%*	*61.5%*	*49.4%*	*35.6%*

Calibration. The AUC measures ranking, not usable detection at a fixed false-positive rate. Table 9 reports both. With the theoretical threshold $\tau^{\rm th}$, the noise-free-trained detector flags 18–49% of other-code spectra at 10–50 ppm. However, it also flags 6.6–14.0% of own-code spectra instead of the nominal 1%. The excess comes from the upper tail of the autoencoder's reconstruction error, which is large compared with σ for the hardest planets. With the empirical threshold $\tau^{\rm emp}$, the realised own-code false-positive rate is 0.8%, close to nominal. At this threshold, however, only 4.1–4.2% of other-code spectra are flagged, roughly independent of σ. The oracle, at its nominal 1% rate, flags 36–81% of other-code spectra. This contrast locates the limitation: the data contain substantial separable signal at 10–50 ppm, but the one-class detector's worst-case reconstruction error masks it.

Ablation: denoising objective. Training with noisy inputs and clean targets (Eq. 18) does not improve detection. The denoising models have a lower AUC than the noise-free models at every σ (Table 8), and a lower true-positive rate at the empirical threshold (Table 9). Their own-code false-positive rate under $\tau^{\rm th}$ is about 1.6× higher. Ten of the twelve denoising runs met the early-stopping criterion, between 226 and 885 epochs, because the noisy validation loss flattens early. Stopping early leaves the hardest spectra less well reconstructed, which widens the nominal tail that sets the threshold. A denoising objective trades worst-case accuracy for average-case accuracy. For a threshold-based detector this is the wrong trade, because the threshold is set by the worst cases.

Supervised versus one-class. The three-class random forest separates the codes far better than any autoencoder score at every noise level (Table 8). Its accuracy is 0.968, 0.783, 0.684, 0.617, and 0.549 at $\sigma = 0$, 10, 20, 30, and 50 ppm (chance = 0.333). At $\sigma \geq 20$ ppm its pairwise AUC is slightly above the oracle's. The two methods answer different questions. The oracle is a one-class goodness-of-fit test: it asks whether a spectrum is consistent with code c, and it is optimal only against an unspecified alternative. The classifier sees all three codes during training and learns the specific directions $\mathbf{\Delta}^{(ab)}$ along which they differ. It can therefore project out noise orthogonal to

Table 10: Random-forest accuracy for test planets whose mean inter-code residual $\bar{\rho}$ exceeds a threshold. "All": overall accuracy.

$\bar{\rho}$	Share	$\sigma = 0$	10	20	30	50 ppm
All	100%	0.968	0.783	0.684	0.617	0.549
≥ 10 ppm	51.0%	0.967	0.905	0.835	0.769	0.687
≥ 30 ppm	23.6%	0.953	0.924	0.903	0.867	0.809
≥ 50 ppm	15.8%	0.948	0.924	0.912	0.885	0.846
≥ 100 ppm	8.6%	0.932	0.913	0.909	0.882	0.851

those directions. The reconstruction objective of the autoencoder contains no information about these directions. The classifier's inductive bias, supervision on the alternative hypotheses, matches the detection task; the autoencoder's does not. petitRADTRANS is the hardest class at every σ, with recall 0.95 at 0 ppm and 0.50 at 50 ppm. pRT−TRX is the closest pair in Table 6, with the smallest offset and the smallest ρ. This does not explain why petitRADTRANS is harder to identify than TauREx, which has a recall of 0.98 at 0 ppm and 0.54 at 50 ppm. The training accuracy is 1.000 at every σ, so the forest overfits. Its test performance is a lower bound on that of a regularised classifier.

Which planets are detectable. Classifier accuracy depends mainly on the amplitude of the inter-code signal (Table 10). At $\sigma = 50$ ppm, planets with mean residual $\bar{\rho} \geq 30$ ppm (23.6% of the test set) are attributed correctly 80.9% of the time, against 54.9% for all planets. Under noise, the most detectable planets are the low-gravity, large-feature planets, which produce the largest absolute differences between codes. At 50 ppm, accuracy is 0.72 in the lowest and 0.45 in the highest octile of $\log g$, and 0.37 in the lowest and 0.81 in the highest octile of feature height.

Planets that limit the autoencoders. The worst-reconstructed 5% of own-code held-out spectra (658 of 13,166 planets) are strongly concentrated in two regions of parameter space:

- cool atmospheres: median T of 768 K versus 1940 K for the remaining planets in POSEIDON, with $A_T = 0.16$, 0.17, and 0.33 for POSEIDON, petitRADTRANS, and TauREx;
- low surface gravity, with $A_{\log g} = 0.24$, 0.22, and 0.17, and correspondingly large H/R_p ($A = 0.59$, 0.62, 0.80).

The feature height has no comparable effect ($A_h = 0.52$–0.66), so these are not spectra near the selection threshold. The poorly reconstructed sets largely coincide across codes. POSEIDON and petitRADTRANS share 479 of 658 planets, where 33 are expected by chance, and the Spearman correlation of their per-planet errors is 0.92. 291 planets are poorly reconstructed in all three codes. The difficulty is therefore a property of the planet, not of the code. Both regions are sparsely populated after selection. The feature-height cut retains 18.6% of planets with $T < 600$ K, against 55.1% with $T > 2000$ K. Only 2,092 selected planets (4.8%) have $\log g < -0.9$, the low-mass edge of the M_p prior. These same low-gravity planets are the easiest to classify under noise. Their spectra are large and structured, which makes the code differences large in absolute terms, but also hard to compress into eight dimensions.

Table 11: XGBoost retrieval R^2 on test planets. Self: mean over the three codes ($b = a$). Cross: mean over the six pairs ($b \neq a$). Gap: self minus cross. Models are trained without noise; the noisy rows apply the same models to test spectra with white noise. In the gap rows, bold marks the largest and underline the second-largest self–cross gap.

σ		T	$\log X_{CO_2}$	$\log X_{CH_4}$	$\log X_{H_2O}$	M_p	$\log X_{NH_3}$	$\log X_{CO}$	$R_\star$	R_p
0 ppm	Self	0.997	0.994	0.966	0.959	0.901	0.901	0.851	0.761	0.609
	Cross	0.963	0.983	0.951	0.953	0.866	0.873	0.627	0.661	0.437
	Gap	0.034	0.011	0.015	0.006	0.035	0.028	**0.224**	0.100	<u>0.172</u>
10 ppm	Self	0.962	0.964	0.900	0.888	0.856	0.790	0.596	0.578	0.349
	Cross	0.931	0.954	0.891	0.884	0.834	0.774	0.438	0.525	0.250
	Gap	0.031	0.010	0.009	0.004	0.022	0.016	**0.158**	0.053	<u>0.099</u>
50 ppm	Self	0.678	0.769	0.680	0.541	0.608	0.296	−0.459	−0.010	−0.572
	Cross	0.659	0.764	0.674	0.538	0.600	0.291	−0.533	−0.015	−0.598

5.4 Q3: Retrieval bias from code mismatch

Table 11 compares self- and cross-retrieval. Without noise, self-retrieval recovers T, $\log_{10} X_{CO_2}$, $\log_{10} X_{CH_4}$, and $\log_{10} X_{H_2O}$ with $R^2 = 0.959$–0.997. R_p and $R_\star$ are only partly constrained ($R^2 = 0.61$ and 0.76), because the transit depth fixes mainly the ratio $(R_p/R_\star)^2$. Replacing the test code reduces R^2 for every parameter. The largest losses are for $\log_{10} X_{CO}$ ($0.851 \to 0.627$) and R_p ($0.609 \to 0.437$). The smallest are for the water, carbon dioxide, and methane abundances (≤ 0.015). The worst single cross-retrieval is R_p from TauREx spectra with a POSEIDON-trained model ($R^2 = 0.238$).

Bias structure. Table 12 lists the median biases for the most affected parameters. Self-retrieval biases are negligible ($|\beta_T| < 0.4\,\mathrm{K}$, $|\beta_{CO}| \leq 0.03\,\mathrm{dex}$). Cross-retrieval biases are largest whenever POSEIDON is involved. pRT↔TRX biases are small: at most 9 K in T and 0.05 dex in CO. This pattern matches the residual structure of Table 6, where the two pairs involving POSEIDON have the largest offsets and short-wavelength differences. The temperature bias depends on temperature. For petitRADTRANS-trained models applied to POSEIDON spectra it grows from $-19\,\mathrm{K}$ at $T \approx 760\,\mathrm{K}$ to $-272\,\mathrm{K}$ at $T \approx 2830\,\mathrm{K}$ (median per temperature sextile), and reaches $-330\,\mathrm{K}$ in the hottest 5% of planets. The R_p and $R_\star$ biases have the same sign in all six pairs. This is consistent with partial compensation along the R_p–$R_\star$ degeneracy of the transit depth. The CO abundance is the most code-sensitive parameter in relative terms, with biases of 3.6–6.2% of its prior range whenever POSEIDON is involved. CO has the weakest self-retrieval constraint among the five molecules ($R^2 = 0.85$). For a weakly constrained parameter, a small systematic change in the spectrum maps onto a large change in the estimate.

Ablation: training-set size. More training data widens the gap between self- and cross-retrieval instead of closing it. From 5% to 100% of the training planets, the self-retrieval R^2 for CO rises from 0.653 to 0.851, while the cross-retrieval R^2 rises only from 0.547 to 0.627; the gap doubles from 0.106 to 0.224. For R_p the gap grows from 0.006 to 0.172. The cross-retrieval R^2 peaks at 50% of the data (0.460) and declines with the full set (0.437). Cross-retrieval of T is flat at 0.950–0.963 over the whole range, while self-retrieval improves from 0.978 to 0.997. A more accurate emulator of one code is a more specialised one, so the code systematics set a floor on cross-code accuracy that additional simulations of a single code do not lower.

Table 12: Median retrieval bias β (predicted − true) on noise-free test spectra for the most code-sensitive parameters. Self-retrieval biases are $|\beta_T| < 0.4\,\mathrm{K}$, $|\beta_{\mathrm{CO}}| \leq 0.03\,\mathrm{dex}$, $|\beta_{R_p}| \leq 0.012\,R_\mathrm{J}$, and $|\beta_{R_\star}| \leq 0.011\,R_\odot$.

Trained → tested	T (K)	$\log X_{\mathrm{CO}}$ (dex)	R_p (R_J)	$R_\star$ ($R_\odot$)
PSN → pRT	+39.1	+0.29	−0.078	−0.052
PSN → TRX	+38.6	+0.22	−0.044	−0.025
pRT → PSN	−45.0	−0.29	+0.056	+0.038
pRT → TRX	−1.9	+0.03	+0.021	+0.017
TRX → PSN	−27.5	−0.37	+0.039	+0.027
TRX → pRT	+9.1	−0.05	−0.018	−0.021

Interaction with noise. Noise reduces R^2 for all parameters and closes most self–cross gaps (Table 11). At 50 ppm the cross-retrieval R^2 is negative for R_p, $R_\star$, and CO. These models were not trained with noise, so the noisy-test values are a lower bound. For PSN→pRT the code biases are nearly unchanged at 10 ppm: T goes from +39 to +38 K, CO from +0.29 to +0.26 dex, and R_p from −0.078 to −0.070 R_J. Where the noise-induced shift has the same sign as the code bias, the two add: for pRT→PSN, β_T grows from −45 to −63 K. The scatter, by contrast, grows: the self-retrieval half-width w_T rises from 27 to about 86 K at 10 ppm and 305 K at 50 ppm. The median ratio $|\beta|/w$ over all cross pairs and parameters therefore falls from 0.10 without noise to 0.08 at 50 ppm. The largest ratios are for T: up to 0.72 without noise and 0.39 at 50 ppm. For an individual planet observed at $\gtrsim 30$ ppm precision, the code systematic is a minor term in the error budget. For a population analysis, where the scatter averages down as $1/\sqrt{N}$ but the bias does not, it remains a coherent shift [TODO_population_level_retrieval_biases].

5.5 Computational cost

Table 13 summarises model size, training cost, and the best realisable detection AUC at $\sigma = 20$ ppm. Within the autoencoder family, AE-W costs 3.2× the parameters and multiply–accumulate operations of AE-S, and 1.7× the training time per 1000 epochs (1123 versus 651 s). It lowers the test MSE by 1.7× (Table 3). Inference costs about 1.1×10^5 multiply–accumulate operations per spectrum, which is negligible for any survey-sized sample [TODO: VERIFY measured inference throughput]. Across model families, the random forest is both cheaper to train (66–70 s per noise level with four threads, against about 10^3 s per autoencoder with one thread) and more accurate at every $\sigma > 0$. For detecting code systematics, the additional cost of the one-class autoencoders brings no benefit. Their value in this study lies in characterising reconstruction difficulty across parameter space (Section 5.3), not in detection.

6 Limitations

Simplified atmospheres. All spectra assume an isothermal, cloud-free atmosphere, and the codes use their default opacity databases. Clouds, hazes, non-isothermal profiles, and disequilibrium chemistry would introduce further modelling choices on which the codes may disagree. Inter-code differences for realistic atmospheres may therefore be larger than those reported here. The CH_4 and NH_3 abundances are restricted above 2000 K and 1600 K because opacity data are unavailable there, which leaves part of the hot regime sparsely covered for those species [TODO: VERIFY thresholds].

Table 13: Cost and detection performance. Times are wall-clock on CPU [TODO: VERIFY processor]. AUC: mean over code pairs at $\sigma = 20$ ppm. For XGBoost, the time per regressor includes four learning-curve refits. Bold: best; underline: second best.

Model	Size	Training (threads)	AUC_{20}
AE-W (×3 codes)	114,972 params	920–1069 s (1)	<u>0.669</u>
Denoising AE-W (×3)	114,972 params	217–954 s (1)	0.630
Random forest	300 trees	66–70 s per σ (4)	**0.886**
XGBoost (×27, retrieval)	$\leq$ 3000 trees, depth 8	80 s mean (4)	–

Confounded sources of disagreement. The three codes differ in opacity data, in opacity treatment (correlated-k at $R = 1000$ for petitRADTRANS, cross-sections at $R = 15{,}000$ for the other two), and in internal conventions. Our paired design measures their combined effect. It does not attribute a residual to a single cause, and the physical interpretations in Section 5.2 are hypotheses that require controlled re-simulation to test.

Noise model. Observational noise is white, Gaussian, and uniform across bins. Real Ariel and JWST errors vary with wavelength and include correlated instrumental and stellar systematics, which can either mask or mimic inter-code differences.

Sample selection. The 210 ppm feature-height cut retains 44.6% of planets and preferentially removes cool planets (18.6% retained below 600 K, 55.1% above 2000 K). Population-level statistics therefore describe large-feature planets, and the hardest planets for the autoencoders are underrepresented in training.

Model and statistical scope. Retrievals are point estimates from gradient-boosted trees trained without noise, not Bayesian posteriors. Most of these regressors reached the cap of 3000 boosting rounds, so they are not fully converged. The random-forest classifier fits its training set perfectly. Results use a single train/validation/test split, two to three initialisation seeds per autoencoder configuration, and one seed for the TauREx detector. The width–schedule ablation varies the learning-rate schedule and the early-stopping patience jointly. Ten of the twelve denoising models stopped early, so the negative result for the denoising objective may partly reflect the stopping rule rather than the objective.

7 Conclusion

We measured the systematic differences between three widely used radiative-transfer codes on a paired sample of 98,725 simulated planets, and traced their consequences for detection and for machine-learning retrieval. The differences consist of a depth offset of about 20 ppm between POSEIDON and the other two codes, which scales with feature height, and a shape residual of about 1% of the feature height. The shape residual is concentrated in a few spectral regions and is a near-deterministic function of the planet parameters. Without noise these differences are easily detected, even when they are only a few ppm. At realistic noise levels they become difficult to detect from a single spectrum, and the difficulty is set by the detector: one-class autoencoders are limited by their own worst-case reconstruction errors, whereas a classifier trained on all three codes

approaches or exceeds a parameter-aware oracle. In retrieval, code mismatch produces parameter-dependent biases that additional training data from a single code do not remove. For individual noisy spectra these biases are small compared with the statistical scatter, but for population studies they persist as coherent shifts.

These results suggest four practices. First, ML retrievals should be validated on spectra from forward models other than the one used for training, and should report the resulting self/cross gap. Second, because the residuals are low-dimensional and predictable from the parameters, they could be emulated and either corrected or included as a modelled error term; we have not tested this here. Third, code identification should use discriminative models trained on all candidate codes rather than one-class reconstruction scores. Fourth, reconstruction-based detectors should be evaluated at a calibrated false-positive rate, because their AUC can greatly overstate their usable detection rate. Extending the paired design to cloudy, non-isothermal atmospheres, shared opacity sources, and instrument-specific noise would separate the physical origins of the systematics and test whether these conclusions hold for real observations.